

Philip (Flip) Tanedo [he/him]

CONTACT INFORMATION

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RESEARCH

I am a theoretical physicist seeking to understand what dark matter is, how it got here, why it is still here, and what we have to do to confirm its identity as an elementary particle. I use quantum field theory to address open questions in fundamental science.

EMPLOYMENT

 **University of California, Riverside**
Associate Professor, Department of Physics & Astronomy 2022 –
Equity Advisor, College of Agricultural and Natural Sciences 2022 –
Assistant Professor, Department of Physics and Astronomy 2016 – 2022

 **University of California, Irvine**
Postdoctoral Scholar, Chancellor's ADVANCE Postdoctoral Fellow 2013 – 2016

EDUCATION

 **Cornell University**, Ithaca, New York
Advisor: Csaba Csáki; *Doctor of Philosophy (2013)*, *Master of Science (2010)* 2008 – 2013

 **Durham University**, Durham, UK
Advisor: Athanasios Dedes; *Master of Science by Research* 2007 – 2008

 **Cambridge University**, Cambridge, UK
Master of Advanced Study in Mathematics with Merit 2006 – 2007

 **Stanford University**, Stanford, CA
Bachelor of Science in Physics and Mathematics with Honors and Distinction 2002 – 2006

FELLOWSHIPS & HONORS

Adviser of the Year 2023
Awarded by the UCR Office of Student Life to a UCR faculty member for outstanding mentorship of a student group. Nominated by the Physics Organization for Womxn and the Under-Represented. One award per year.

National Science Foundation CAREER Award 2021 – 2026
The NSF's most prestigious award for early-career faculty with the potential to serve as academic role models in research and education and to lead advances in the mission of their department. Approximately one award in theoretical high-energy physics per year.

UCR Junior Faculty Excellence in Teaching Award 2021
The UCR Academy of Distinguished Teaching award for junior faculty. Up to 3 awards per year.

Hellman Fellowship 2020 – 2021
Awarded by the Hellman Fellows Fund to support to junior faculty who show capacity for great distinction in their research. 7 UCR fellows per year.

Commitment to Graduate Diversity Award 2020
Awarded by UCR Graduate Division to up to two faculty per year who champion efforts to promote a diverse and inclusive graduate student and campus climate.

UCI Chancellor's ADVANCE Postdoctoral Fellowship 2014 – 2015
Fellowship for postdoctoral scholars with strong promise for faculty appointment and a commitment to equal opportunity. Drawn from finalists of the UC President's Postdoctoral Fellowship Program.

Ford Foundation Dissertation Fellowship, Honorable Mention 2012
Individuals with evidence of superior academic achievement and committed to a career in teaching and research at the college or university level in the U.S.

Paul and Daisy Soros Fellowship for New Americans

2010 – 2012

For first- and second-generation immigrants poised to make significant contributions to society, culture or their academic field. 30 fellows per year.

Ford Foundation Predoctoral Fellowship, Honorable Mention

2010

Individuals with evidence of superior academic achievement and committed to a career in teaching and research at the college or university level in the U.S.

National Science Foundation Graduate Research Fellowship

2006

Outstanding graduate students who have demonstrated the potential to be high achieving scientists and engineers, early in their careers. Approximately 2000 awards out of 12,000 applicants across the country and across disciplines.

Marshall Scholarship

2006

Awarded by the UK Foreign, Commonwealth and Development Office to Americans of high ability to pursue graduate study in the UK. 50 scholars per year out of a pool of 1000 university-endorsed applicants.

Barry M. Goldwater Scholarship

2005

The most prestigious undergraduate scholarship in the natural sciences, mathematics, and engineering in America. Approximately 400 awards out of 5000 2nd and 3rd year undergraduates; two nominees per institution.

PUBLICATIONS

The standard practice in particle physics is to list authors in alphabetical order.

“Holography of broken $U(1)$ symmetry”

I. Chaffey, P. Tanedo, S. Fichet [arXiv:2309.00040](#)
JHEP **05**, 330 (2024) [doi:10.1007/JHEP05\(2024\)330](#)

“Snowmass 2021 Dark Matter Complementarity Report”

A. Boveia + 26 others [arXiv:2211.07027](#)

“Snowmass Theory Frontier Report”

N. Craig + 41 others [arXiv:2211.05772](#)

“Snowmass 2021 Cross Frontier Report: Dark Matter Complementarity (Extended Version)”

A. Boveia + 26 others [arXiv:2210.01770](#)

“Snowmass Theory Frontier: Astrophysics and Cosmology”

D. Green + 86 others [arXiv:2209.06854](#)

“Dark Z at the International Linear Collider”

M. Perelstein, P. Tanedo, Y. San [arXiv:2205.10304](#)
Phys. Rev. D **106**, 015027 (2022) [doi:10.1103/PhysRevD.106.015027](#)

“The International Linear Collider: Report to Snowmass 2021”

A. Aryshev + 510 others [arXiv:2203.07622](#)

“Efficient sampling of constrained high-dimensional theoretical spaces with machine learning”

D. Whiteson, J. Hollingsworth, M. Ratz, P. Tanedo [arXiv:2103.06957](#)
Eur. Phys. J. C **81**, 1138 (2021) [doi:10.1140/epjc/s10052-021-09941-9](#)

“Continuum-Mediated Self-Interacting Dark Matter”

I. Chaffey, P. Tanedo, S. Fichet [arXiv:2102.05674](#)
JHEP **06**, 008 (2021) [doi:10.1007/JHEP06\(2021\)008](#)

“Dark kinetic heating of neutron stars from contact interactions with relativistic targets”

A. Joglekar, H. Yu, N. Raj, P. Tanedo [arXiv:2004.09539](#)
Phys. Rev. D **102**, 123002 (2020) [doi:10.1103/PhysRevD.102.123002](#)

“Effective Field Theory in AdS: Continuum Regime, Soft Bombs, and IR Emergence”

A. Costantino, P. Tanedo, S. Fichet [arXiv:2002.12335](#)
Phys. Rev. D **102**, 115038 (2020) [doi:10.1103/PhysRevD.102.115038](#)

“Relativistic capture of dark matter by electrons in neutron stars”

A. Joglekar, H. Yu, N. Raj, P. Tanedo [arXiv:1911.13293](#)
Phys. Lett. B **809**, 135767 (2020) [doi:10.1016/j.physletb.2020.135767](#)

“Exotic Lepton-Flavor Violating Higgs Decays”

J. Evans, M. Zakeri, P. Tanedo [arXiv:1910.07533](#)
JHEP **01**, 028 (2020) [doi:10.1007/JHEP01\(2020\)028](#)

- “Exotic Spin-Dependent Forces from a Hidden Sector”
A. Costantino, P. Tanedo, S. Fichet [arXiv:1910.02972](#)
JHEP **03**, 148 (2020) [doi:10.1007/JHEP03\(2020\)148](#)
- “Vector self-interacting dark matter”
I. Chaffey, P. Tanedo [arXiv:1907.10217](#)
Phys. Rev. D **101**, 075005 (2020) [doi:10.1103/PhysRevD.101.075005](#)
- “The Warped Dark Sector”
P. Brax, P. Tanedo, S. Fichet [arXiv:1906.02199](#)
Phys. Lett. B **798**, 135012 (2019) [doi:10.1016/j.physletb.2019.135012](#)
- “DarkCapPy: Dark Matter Capture and Annihilation”
A. Green, P. Tanedo [arXiv:1808.03700](#)
Comput. Phys. Commun. **242**, 120 (2019) [doi:10.1016/j.cpc.2019.04.010](#)
- “Just a taste: lectures on flavor physics.”
P. Tanedo, Y. Grossman [arXiv:1711.03624](#) [doi:10.1142/9789813233348_0004](#)
- “Neutron stars at the dark matter direct detection frontier”
H. Yu, N. Raj, P. Tanedo [arXiv:1707.09442](#)
Phys. Rev. D **97**, 043006 (2018) [doi:10.1103/PhysRevD.97.043006](#)
- “US Cosmic Visions: New Ideas in Dark Matter 2017: Community Report”
M. Battaglieri + 250 others [arXiv:1707.04591](#)
- “Dark Photons from Captured Inelastic Dark Matter Annihilation: Charged Particle Signatures”
J. Smolinsky, P. Tanedo [arXiv:1701.03168](#)
Phys. Rev. D **95**, 075015 (2017) [doi:10.1103/PhysRevD.95.075015](#)
- “Dark Matter Interpretation of the Fermi-LAT Observation Toward the Galactic Center”
C. Karwin, P. Tanedo, S. Murgia, T. Porter, T. Tait [arXiv:1612.05687](#)
Phys. Rev. D **95**, 103005 (2017) [doi:10.1103/PhysRevD.95.103005](#)
- “Lepton-Flavor Violating Mediators”
A. Kwa, I. Galon, P. Tanedo [arXiv:1610.08060](#)
JHEP **03**, 064 (2017) [doi:10.1007/JHEP03\(2017\)064](#)
- “Dark Sectors 2016 Workshop: Community Report”
J. Alexander + 203 others [arXiv:1608.08632](#)
- “Particle physics models for the 17 MeV anomaly in beryllium nuclear decays”
B. Fornal, I. Galon, J. Feng, J. Smolinsky, P. Tanedo, S. Gardner, T. Tait [arXiv:1608.03591](#)
Phys. Rev. D **95**, 035017 (2017) [doi:10.1103/PhysRevD.95.035017](#)
- “Protophobic Fifth-Force Interpretation of the Observed Anomaly in ^8Be Nuclear Transitions”
B. Fornal, I. Galon, J. Feng, J. Smolinsky, P. Tanedo, S. Gardner, T. Tait [arXiv:1604.07411](#)
Phys. Rev. Lett. **117**, 071803 (2016) [doi:10.1103/PhysRevLett.117.071803](#)
- “Kaluza-Klein gluons at 100 TeV: NLO corrections”
B. Lillard, P. Tanedo, T. Tait [arXiv:1602.08622](#)
Phys. Rev. D **94**, 054012 (2016) [doi:10.1103/PhysRevD.94.054012](#)
- “Beyond the Standard Model”
C. Csáki, P. Tanedo [arXiv:1602.04228](#) [doi:10.5170/CERN-2015-004.169](#)
- “Detecting dark matter through dark photons from the Sun: Charged particle signatures”
J. Feng, J. Smolinsky, P. Tanedo [arXiv:1602.01465](#)
Phys. Rev. D **93**, 115036 (2016) [doi:10.1103/PhysRevD.93.115036](#)
- “Limiting SUSY compressed spectra scenarios”
A. Nelson, D. Whiteson, P. Tanedo [arXiv:1509.08485](#)
Phys. Rev. D **93**, 115029 (2016) [doi:10.1103/PhysRevD.93.115029](#)
- “Dark Photons from the Center of the Earth: Smoking-Gun Signals of Dark Matter”
J. Feng, J. Smolinsky, P. Tanedo [arXiv:1509.07525](#)
Phys. Rev. D **93**, 015014 (2016) [doi:10.1103/PhysRevD.93.015014](#)

- “Dynamics of 3D SUSY Gauge Theories with Antisymmetric Matter”
C. Csáki, J. Terning, M. Martone, P. Tanedo, Y. Shirman [arXiv:1406.6684](#)
JHEP **08**, 141 (2014) [doi:10.1007/JHEP08\(2014\)141](#)
- “Hidden on-shell mediators for the Galactic Center γ -ray excess”
A. DiFranzo, A. Rajaraman, A. Wijangco, M. Abdullah, P. Tanedo, T. Tait [arXiv:1404.6528](#)
Phys. Rev. D **90**, 035004 (2014) [doi:10.1103/PhysRevD.90.035004](#)
- “Effective theory of self-interacting dark matter”
B. Bellazzini, M. Cliche, P. Tanedo [arXiv:1307.1129](#)
Phys. Rev. D **88**, 083506 (2013) [doi:10.1103/PhysRevD.88.083506](#)
- “The Same-Sign Dilepton Signature of RPV/MFV SUSY”
J. Berger, M. Perelstein, M. Saelim, P. Tanedo [arXiv:1302.2146](#)
JHEP **04**, 077 (2013) [doi:10.1007/JHEP04\(2013\)077](#)
- “Fundamental Physics at the Intensity Frontier”
J. Hewett + 467 others [arXiv:1205.2671](#) [doi:10.2172/1042577](#)
- “The Birds and the B s in RS: The $b \rightarrow s\gamma$ penguin in a warped extra dimension”
B. Shakya, M. Blanke, P. Tanedo, Y. Tsai [arXiv:1203.6650](#)
JHEP **08**, 038 (2012) [doi:10.1007/JHEP08\(2012\)038](#)
- “SUSY_FLAVOR v2: A Computational tool for FCNC and CP-violating processes in the MSSM”
A. Crivellin, A. Dedes, J. Rosiek, P. Chankowski, P. Tanedo, S. Jaeger [arXiv:1203.5023](#)
Comput. Phys. Commun. **184**, 1004 (2013) [doi:10.1016/j.cpc.2012.11.007](#)
- “Goldstone Fermion Dark Matter”
B. Bellazzini, C. Csaki, J. Hubisz, J. Shao, P. Tanedo [arXiv:1106.2162](#)
JHEP **09**, 035 (2011) [doi:10.1007/JHEP09\(2011\)035](#)
- “Warped penguin diagrams”
C. Csaki, P. Tanedo, Y. Grossman, Y. Tsai [arXiv:1004.2037](#)
Phys. Rev. D **83**, 073002 (2011) [doi:10.1103/PhysRevD.83.073002](#)
- “SUSY_FLAVOR: A Computational Tool for FCNC and CP-violating Processes in the MSSM”
A. Dedes, J. Rosiek, P. Chankowski, P. Tanedo, S. Jager [arXiv:1003.4260](#)
Comput. Phys. Commun. **181**, 2180 (2010) [doi:10.1016/j.cpc.2010.07.047](#)
- “Complete One-Loop MSSM Predictions for $B \rightarrow \ell\ell'$ at the Tevatron and LHC”
A. Dedes, J. Rosiek, P. Tanedo [arXiv:0812.4320](#)
Phys. Rev. D **79**, 055006 (2009) [doi:10.1103/PhysRevD.79.055006](#)

INVITED TALKS

<i>Astrophysics Tests of Dark Sectors</i>	
Invited Plenary, Dark Interactions 2024, Vancouver, Canada	Dec. 2024
<i>Direct Collapse Black Holes from Dark Matter</i>	
Invited Plenary, Lake Louise Winter Institute 2025, Canada	Feb. 2025
Chalkboard Talk, Kavli Institute for Theoretical Physics, Santa Barbara, CA	Jul. 2024
Invited Talk, Center for Theoretical Underground Physics and Related Areas, SD	Jun. 2024
Invited Plenary, Mitchell Conference, College Station, TX	May. 2024
Invited Plenary, Dark Matter, First Light Workshop, Perimeter Institute, Canada	Feb. 2024
<i>Why have we not [yet?] discovered dark matter?</i>	
Invited Colloquium, University of Oregon, OR	Feb. 2023
Invited Colloquium, Carleton University, Canada	Dec. 2023
Invited Colloquium, UCSB, CA	Nov. 2023
Invited Hybrid Seminar, UW Bothell Physics Seminar, Bothell, WA	Oct. 2023
Invited Hybrid Seminar, UW Bothell Physics Seminar, Bothell, WA	Nov. 2022
Invited Seminar, Loyola University, Chicago, IL	Sep. 2022
Invited Colloquium, California State University, Long Beach CA	Sep. 2023
<i>Adventures in a Warped Dark Sector</i>	

Invited Physics Seminar, Carleton University, Canada	Dec. 2023
Invited Physics Seminar, University of Melbourne (Hybrid), Australia	Oct. 2022
<i>Neutron Star Heating from Dark Matter: The Relativistic Frontier</i>	
Invited Talk, Center for Theoretical Underground Physics and Related Areas, SD	Jul. 2023
<i>Introduction to Hidden Sectors</i>	
Invited Overview Talk, YOUNGST@RS workshop (hybrid), MITP, Mainz	Nov. 2023
<i>Adventures in a Warped Dark Sector</i>	
Invited Physics Seminar, University of Melbourne (Hybrid), Australia	Oct. 2022
<i>Dark Matter in Compact Objects</i>	
Invited Plenary Talk, SAIIR/Princípia Workshop on the Nature of Dark Matter, São Paulo, Brazil	Nov. 2022
<i>Theory and the Cosmic Frontier</i>	
Invited Talk, KITP Conference: Snowmass Theory Frontier, Santa Barbara, CA	Feb. 2021
<i>Some [likely] Wrong Ideas About Dark Matter and What They Teach Us</i>	
Invited Colloquium, Colloquium, UCR Physics & Astronomy Department, Riverside, CA	Oct. 2021
<i>The Invisible Universe and You</i>	
Invited Talk, UCR Hellman Fellows Symposium, Riverside, CA	Mar. 2021
<i>Warped Dark Sectors</i>	
Invited Talk, A Rainbow of Dark Sectors, Aspen Center for Physics, CO	Mar. 2021
Invited Seminar, Korea Institute for Advanced Studies, Seoul, Korea	Nov. 2020
<i>e^+e^- Production of Dark Sector Particles</i>	
Invited Talk, Americas Workshop on Linear Colliders 2020, SLAC, Menlo Park, CA	Oct. 2020
<i>Dark Matter Heating Neutron Stars: the Electronic Frontier</i>	
Invited talk, Particle Seminar, Brookhaven National Lab, NY	Feb. 2020
<i>How Not to Discover the WIMP / Whatever Happened to the WIMP of Tomorrow?</i>	
Invited Colloquium, Cornell Physics Department, Ithaca, NY	Oct. 2020
Colloquium, UCR Physics & Astronomy Department, Riverside, CA	Oct. 2020
Invited Colloquium, Aspen Center for Physics, CO	Jun. 2019
Invited Plenary, Samahang Pisika ng Pilipinas Annual Conference, Philippines	May. 2019
Invited Seminar, National Institute for Physics/UP Diliman, Philippines	May. 2019
Invited Seminar, Sam Houston State University, Huntsville, TX	Apr. 2019
Invited Student Selected Physics Colloquium, University of Arizona, Tucson, AZ	Apr. 2019
Invited Seminar, University of Louisiana Physics Department, Lafayette, LA	Apr. 2019
<i>Vector Self-Interacting Dark Matter</i>	
Invited Talk, 2019 Mitchell Conference, College Station, TX	May. 2019
Invited Seminar, Texas A&M Mitchell Institute HEP Seminar, College Station, TX	Apr. 2019
<i>Dark Matter Complementarity (Is the WIMP Dead?)</i>	
Invited Colloquium, Harvey Mudd Physics Colloquium, Pomona, CA	Nov. 2018
Invited Colloquium, USC, Los Angeles, CA	Oct. 2018
Invited Ad-Hoc Talk, The Small-Scale Structure of Cold (?) Dark Matter Workshop, KITP, Santa Barbara, CA	Jun. 2018
<i>Composite Mediators</i>	
Invited Plenary Talk, Santa Fe Workshop: Trending Topics in HEP, Santa Fe, NM	Jul. 2018
Invited Talk, Matter Detection and Detectability: Paradigm Confirmation or Shift? KITP, Santa Barbara, CA	Apr. 2018
Invited Seminar, Lawrence Berkeley National Laboratory, Berkeley, CA	Apr. 2018
Parallel Talk, American Physical Society April Meeting, Columbus, OH	Apr. 2018
Invited Seminar, Los Alamos National Laboratory, Los Alamos, NM	Mar. 2018

Light Mediators to Dark Sectors

DNP Invited Talk, American Physical Society April Meeting, Columbus, OH

Apr. 2018

Light mediators and terrestrial dark matter capture

Invited Talk, New Directions in Dark Matter and Neutrino Physics Workshop,
Perimeter Institute, Waterloo, CA

Jul. 2017

Models, Exotic Signatures and Proposed Searches

Invited Plenary Talk, XI International Conference on the Interconnection between
Particle Physics and Cosmology, Corpus Christi, TX

May 2017

Composite Higgs, Composite Mediators

Invited Plenary Talk, DM at LHC Conference, UC Irvine, CA

Apr. 2017

New Physics in Nuclear Transitions? The 17 MeV Anomaly in ^8Be

Invited Plenary Talk, Focus Workshop on Particle Physics and Cosmology,
CTPU-IBS, Daejeon City, Korea

Dec. 2016

Invited Plenary Talk, Beyond the Standard Model Workshop, MCTP, Ann Arbor, MI

Oct. 2016

Invited Talk, LHC Awakens: A New Energy Frontier, Aspen Center for Physics, CO

Sep. 2016

Invited Webinars, Latin American Webinars on Physics

Sep. 2016

Plenary Talk, LHC Run II Workshop, Santa Fe, NM

Jul. 2016

Invited Seminar, Cornell Particle Theory Seminar, Ithaca, NY

May 2016

Invited Seminar, UMD Particle Theory Seminar, College Park, MD

May 2016

Invited Talk, UC Davis Theory Lunch, Davis, CA

May 2016

Parallel Talk, Dark Sectors 2016 Workshop, SLAC Menlo Park, CA

Apr. 2016

Dark Sunshine

Invited Physics Colloquium, Cal State Fullerton Physics, Fullerton, CA

Mar. 2016

Invited Physics Colloquium, CSU Long Beach, CA

Mar. 2016

Dark Photons from the Center of the Earth

Invited Seminar, UT Austin Physics Seminar, Austin, TX

Feb. 2017

Invited Seminar, UC Davis Theory Seminar, Davis, CA

May 2016

Invited Plenary Talk, PIKIO Workshop, Cincinnati, OH

Feb. 2016

Invited Seminar, MCTP Brown Bag Seminar, Ann Arbor, MI

Feb. 2016

Invited Seminar, SLAC Theory Seminar, Menlo Park, CA

Feb. 2016

Invited Seminar, UC Riverside Theory Seminar, Riverside, CA

Dec. 2015

Invited Seminar, TRIUMF Theory Seminar, Vancouver, Canada

Dec. 2015

Invited Seminar, UC Santa Barbara HEP Experiment & Phenomenology Seminar

Sep. 2015

Invited talk, New Directions to Shed Light on Dark Matter, Aspen, CO

Sep. 2015

Parallel Talk, SUSY 2015: 23rd Conference on SUSY & Unification, Lake Tahoe, CA

Aug. 2015

Parallel Talk, Anticipating 14 TeV: Insights into matter from the LHC & Beyond,
MIAPP Conference, Munich, Germany

Jul. 2015

On-Shell Mediators / Heavy, Hidden Hooperon

Invited Seminar, Fermilab HEP Theory Seminar, Batavia, IL

Dec. 2015

Invited Seminar, UC Davis HEFTI Seminar

Mar. 2015

Invited Seminar, Santa Cruz Institute for Particle Physics Theory Seminar, UCSC

Mar. 2015

Plenary Talk, Santa Fe Summer Workshop: LHC After the Higgs, Santa Fe, NM

Jul. 2014

Invited Seminar, UCLA HEP Theory Seminar, Los Angeles, CA

Jun. 2014

Invited Seminar, May 2014

Invited Seminar, SLAC Theory Seminar, Menlo Park, CA

May 2014

Invited Seminar, BCTP Seminar, UC Berkeley & LBNL, CA

May 2014

Charged Lepton Flavor Violation in RS

Invited Parallel Talk, International Workshop for Charged Lepton Flavor Violation,
UC Irvine, CA

May 2014

Stealthy Composite Stops

Invited Seminar, Perimeter Institute Particle Physics Seminar, Waterloo, Canada

Nov. 2012

Goldstone Fermion Dark Matter

Invited Seminar, UC Davis HEFTI Seminar	Oct. 2012
Invited Seminar, Harvard Particle Theory Seminar, Cambridge, MA	Feb. 2012
Invited Seminar, UC Irvine Joint Particle Seminar	Jan. 2012
Seminar, Cornell Theory Group Seminar, Ithaca, NY	Sep. 2011
Parallel Talk, SUSY 2011: 19th Conference on SUSY & Unification, Fermilab	Aug. 2011

The Birds and the Bs in RS

Parallel Talk, Phenomenology 2012 Symposium, Pittsburgh, PA	May 2012
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RS@LHC

Contributed Talk, Cornell CMS Future Directions Workshop, Ithaca, NY	Jan. 2012
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Flight of the Warped Penguins

Parallel Talk, Phenomenology 2011 Symposium, Madison, WI	May 2011
Seminar, Cornell Particle Theory Lunch Seminar	Feb. 2011
Seminar, Maryland Theory Seminar, UMD	Mar. 2011
Seminar, DAMTP–Cavendish Phenomenology Seminar, Cambridge, UK	Feb. 2011

The Birds and the Bs: $B_s \rightarrow \mu\mu$ in the MSSM

Contributed Talk, Theoretical Advanced Study Institute 2009, CU Boulder, CO	Jun. 2009
Seminar, Cornell Particle Theory Lunch Seminar	Feb. 2009
Parallel Talk, IOP HEPP: Beauty Physics in the UK, Lancaster, UK	Apr. 2008
Seminar, U. Ioannina Div. Theoretical Physics Journal Club, Greece	Oct. 2008
Contributed Talk, LHCb UK Meeting, Durham, UK	Dec. 2007

Theory vs. Experiment in our generation

Contributed Talk, Beyond Part III Conference, Cambridge, UK	Apr. 2009
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New Physics at CDF?

Seminar, Cornell Collider Phenomenology Journal Club	Nov. 2008
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Split SUSY: How I learned to stop worrying and love fine-tuning

Seminar, Cornell Particle Theory Student Seminar	Jul. 2008
Seminar, Durham CPT Student Seminar	Sep. 2007

Quantum penguins hint at the nature of matter and antimatter

Invited Talk, 2008 Marshall Aid Commemoration Commission Colloquium, London	Apr. 2008
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WTF is a B physics? A talk for string theorists

Contributed Talk, 5th Annual Part III Return Conference, Cambridge, UK	Apr. 2008
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INVITED LECTURES,
PUBLIC TALKS

<i>Why We Have Not Discovered Dark Matter: a Theorist's Apology</i> Public Talk (PIRSA ID: 24020103), Perimeter Institute	Feb. 2024
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Why We Haven't Discovered Dark Matter

W. David Baird Distinguished Lecture Series, Pepperdine University	Sept. 2023
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Little Particles in a Big Universe

26th Fr. Ciriaco Pedrosa, O.P. Memorial Lecture Series and 8th International Symposium on Mathematics and Physics, University of Santo Tomas, Philippines	Nov. 2021
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TRISEP Lectures on BSM Physics

Tri-Institute Summer School on Elementary Particles, TRIUMF, Vancouver	Jul. 2019
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WORKSHOPS &
CONFERENCES

<i>Lake Louise Winter Institute 2025</i> , Lake Louise, Canada	Mar. 2025
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What is Particle Theory?,

KITP, Santa Barbara, CA	Jan. 2025
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Matter Theory, Simulation, and Analysis in the Era of Large Surveys, KITP, Santa Barbara, CA	Jun. 2024
Dark Matter, First Light, Perimeter Institute for Theoretical Physics, Waterloo, Canada	Feb. 2024
YOUNGST@RS—Interacting Dark Sectors in Astrophysics, Cosmology, and the Lab, Mainz Institute for Theoretical Physics, Johannes Gutenberg University	Nov. 2023
CETUP* 2023: Dark Matter, Sanford Underground Research Facility, Lead, SD	Jun. 2023
Theoretical Physics for Machine Learning, Aspen Center for Physics, Aspen, CO	Feb. 2023
SAIFR/Princípia Workshop on the Nature of Dark Matter, Principia Institute, ICTP-SAIFR, São Paulo	Nov. 2023
Sustainable HEP 2nd ed. (remote),, CERN	Sep. 2022
Searching for New Physics,, Aspen Center for Physics, CO	Aug. 2022
IAIFI Summer Workshop, Tufts University, MA	Aug. 2022
IAIFI Ph.D Summer School (student), Tufts University, MA	Aug. 2022
First Pollica Workshop on Dark Matter (organizer), Pollica, Italy	Jun. 2022
American Physical Society April Meeting, New York, NY	Apr. 2022
Neutrinos as a Portal to New Physics and Astrophysics, KITP, Santa Barbara, CA	Mar. 2022
Snowmass Theory Frontier Meeting, KITP, Santa Barbara, CA	Feb. 2022
APS IDEA October Workshop, Remote Meeting, American Physical Society	Oct. 2021
Dark Matter from the Laboratory to the Cosmos, Aspen Center for Physics, CO	Aug. 2021
A Rainbow of Dark Sectors, Aspen Center for Physics, CO	Mar. 2021
Americas Workshop on Linear Colliders 2020, Remote meeting, SLAC, Menlo Park, CA	Jun. 2021
Phenomenology Symposium (session chair), Remote meeting, University of Pittsburgh, PA	May 2021
APS IDEA Network Launch Meeting, Remote meeting, American Physical Society	Jun. 2020
Southern California Beyond the Standard Model, UCI, Irvine, CA	Sep. 2019
KITP Origin of the Vacuum Energy & Electroweak Scales, Santa Barbara, CA	Jun. 2019
Samahang Pisika ng Pilipinas (Philippine Physics Society) Conference, Philippines	Jun. 2019
Mitchell Conference, Texas A&M University, College Station, TX	May 2019
KITP Follow-On Workshop (CDM18), Santa Barbara, CA	Mar. 2019
Southern California Beyond the Standard Model (organizer), UCLA, Los Angeles, CA	Jan. 2019
APS Bridge/National Mentoring Community Conference, Stanford, CA	Nov. 2018
Flavor of New Physics in Collisions, Aspen Center for Physics, CO	Aug. 2018
Santa Fe Workshop: Trending Topics in HEP, Santa Fe, NM	Jul. 2018
American Physical Society April Meeting, Columbus, OH	Apr. 2018
Dark Matter in Southern California IV Symposium, Pasadena, CA	Sep. 2017
New Directions in Dark Matter and Neutrino Physics, Perimeter Institute, Waterloo, Canada	Jul. 2017
Weak Interactions and Neutrinos, UCI, Irvine, CA	Jun. 2017
XI International Conference on Particle Physics and Cosmology, Corpus Christi, TX	May 2017
UC President's Postdoctoral Fellow Spring Retreat, Lake Arrowhead, CA	Apr. 2017
Dark Matter at LHC, UCI, Irvine, CA	Apr. 2017
Southern California Beyond the Standard Model (organizer), UCR, Riverside, CA	Apr. 2017
Beyond WIMPs: from Theory to Detection, Simons Center, Stony Brook, NY	Mar. 2017
Focus Workshop on Particle Physics and Cosmology, CTPU-IBS, Korea	Dec. 2016
Beyond the Standard Model Workshop, MCTP, Ann Arbor, MI	Oct. 2016
The LHC Awakens, Aspen Center for Physics, CO	Aug. 2016
Santa Fe Summer Workshop: LHC Run II, Santa Fe, NM	Jul. 2016
New Accelerators for the 21st Century, KITP, Santa Barbara, CA	Jun. 2016
SLAC Dark Sectors 2016 Workshop, Menlo Park, CA	Apr. 2016
PIKIO Workshop, Cincinnati, OH	Feb. 2016
UCLA Dark Matter 2016, Westwood, CA	Feb. 2016

Science Hack Day, San Francisco, CA	Oct. 2015
UCI Data Science Initiative: Predictive Modeling in Python, Irvine, CA	Oct. 2015
UC PFPF Fall Conference, Oakland, CA	Sep. 2015
New Directions to Shed Light on Dark Matter, Aspen Center for Physics, CO	Sep. 2015
Supersymmetry 2015, Lake Tahoe, CA	Aug. 2015
MIAPP: Anticipating 14 TeV, Munich, Germany	Jul. 2014
UC PFPF Academic Retreat, Lake Arrowhead, CA	Apr. 2015
UCI Data Science Initiative: Software Carpentry, Irvine, CA	Feb. 2015
UC PFPF 30th Anniversary Meeting, Oakland, CA	Oct. 2014
Model Building in the LHC Era, Aspen Center for Physics, CO	Aug. 2014
Santa Fe 2014 Summer Workshop, Santa Fe, NM	Jul. 2014
SLAC Workshop on Physics at a 100 TeV Collider, Menlo Park, CA	Apr. 2014
Dark Matter in Southern California III Symposium, Pasadena, CA	Apr. 2014
Dark Matter in Southern California II Symposium, Pasadena, CA	Sep. 2013
TeV Particle Astrophysics, Irvine, CA	Aug. 2013
Prospects in Theoretical Physics, Princeton, NJ	Jul. 2013
Communicating Science 2013, Cambridge, MA	Jun. 2013
Phenomenology 2012 Symposium, Pittsburgh, PA	May 2012
Fundamental Physics at the Intensity Frontier, Rockville, MD	Dec. 2011
Soros Fellowship for New Americans Fall Conference, OSI, New York	Oct. 2011
Supersymmetry 2011, Fermilab, Batavia, IL	Sep. 2011
First Year of the LHC, KITP, Santa Barbara, CA	Aug. 2011
SUSY Breaking '11, CERN, Geneva, Switzerland	May 2011
Phenomenology 2011 Symposium, Madison, WI	May 2011
Soros Fellowship for New Americans Fall Conference, OSI, New York	Oct. 2010
Emerging problems in particle phenomenology, ITS at CUNY	Apr. 2010
Theoretical Advanced Studies Institute, University of Colorado at Boulder	Jun. 2009
International Workshop on SUSY Breaking, IPPP/Durham University	Apr. 2009
Young Researchers in Mathematics (Beyond Part III), Cambridge University	Apr. 2009
School on the Gauge/Gravity Correspondence, ICTP, Trieste	May 2008
Marshall Colloquium, Association of Commonwealth Universities, London, UK	May 2008
5th Part III Return Conference, Cambridge University	Apr. 2008
IOP HEPP Group Annual Meeting, University of Lancaster	Apr. 2008
Higgs–Maxwell Workshop: EWSB, Royal Society of Edinburgh	Feb. 2008
Young Experimentalists and Theorists Institute, IPPP/Durham University	Jan. 2008
Annual Theory Meeting, IPPP/Durham University	Dec. 2007
UK LHCb Annual Meeting, IPPP/Durham University	Dec. 2007
Cosener’s Forum on Heavy Flavor Physics, Cosener’s House, Abingdon, UK	Jun. 2007

SOFTWARE

DarkCapPy, with A. Green http://github.com/agree019/DarkCapPy	Nov. 2018
SUSY_FLAVOR, with J. Rosiek, A. Crivellin, P. Chankowski, A. Dedes, and S. Jaeger http://www.fuw.edu.pl/susy_flavor/	Mar. 2010

ORGANIZED ACTIVITIES

Scientific Organizer: KITP Teachers’ Conference , Santa Barbara, CA	Feb. 2025
Workshop Co-organizer: Dark Pollica Workshop on Dark Matter, Pollica, Italy	Jun. 2022
Workshop Co-organizer: Communicating Science AIP, College Park, MD	Sep. 2019
Fundamental Physics Innovation Award: Gordon & Betty Moore Foundation/APS	

Hosting Sylvain Fichet (Travel Award), UCR, Riverside, CA	Aug. 2019
Hosting Benjamin Nachmann (Visitor Award), UCR, Riverside, CA	Jan. 2019
Workshop Organizer: Southern California Beyond the Standard Model	
SoCal BSM 2017, UCR, Riverside, CA	Apr. 2017
SoCal BSM 2019, UCLA, Los Angeles, CA	Jan. 2019

OUTREACH ACTIVITIES

Moderator: Physical Science (Phy-Sci) Book Club, Cellar Door Books, Riverside, CA	2017 –
Community book club to read and discuss popular science and technology books.	

OTHER ACTIVITIES

UCR Center for Ideas & Society Academic Book Club	
UCR program for cross-discipline critical inquiry and discussions.	
<i>This Is How They Tell Me The World Ends</i> (organizer, 7 members)	Mar. 2022
<i>White Kids & Klara and the Sun</i> (organizer, 6 members)	Nov. 2021
<i>White Kids</i> (organizer, 6 members)	Nov. 2021
<i>The Problem with Everything</i> (organizer, 6 members)	Feb. 2021
<i>iGen</i> (participant, 7 members)	Oct. 2019
Perry Link Seminar	Apr.–Jun. 2019
"What Are the Intrinsic Rewards of Our Work?" seminar. Cross-discipline discussions between 12 invited faculty.	

TEACHING

UC Riverside, Riverside, CA	
<i>Instructor, Physics 017: Linear Algebra for Physics</i>	Spr. 2025
<i>Instructor, Honors 018: Poetry for Physicists (Ignition Seminar)</i>	Fall 2024
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2024
<i>Instructor, Physics 017: Linear Algebra for Physics</i>	Spr. 2024
<i>Instructor, Physics 165: Particle Physics</i>	Spr. 2024
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2023
<i>Instructor, Physics 017: Linear Algebra for Physics</i>	Spr. 2023
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2022
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2022
<i>Instructor, Physics 017: Linear Algebra for Physics</i>	Spr. 2022
<i>Instructor, Physics 165: Particle Physics</i>	Spr. 2021
<i>Instructor, Physics 017: Linear Algebra for Physics</i>	Spr. 2021
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2021
<i>Instructor, Honors 018: Poetry for Physicists (Ignition Seminar)</i>	Fall 2021
<i>Instructor, Physics 40B: General Physics</i>	Win. 2021
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2020
<i>Instructor, Physics 40B: General Physics</i>	Spr. 2020
<i>Instructor, Physics 230B: Quantum Field Theory</i>	Win. 2020
<i>Instructor, Physics 165: Particle Physics</i>	Win. 2020
<i>Instructor, Honors 018: Poetry for Physicists (Ignition Seminar)</i>	Fall 2019
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2019
<i>Instructor, Physics 262: Special Topics: Group Theory</i>	Win. 2019
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2018
<i>Instructor, Physics 177: Computational Methods for Physical Sciences</i>	Spr. 2018
<i>Instructor, Physics 165: Particle Physics</i>	Win. 2018
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2017
<i>Instructor, Physics 177: Computational Methods for Physical Sciences</i>	Spr. 2017
<i>Instructor, Physics 208: General Relativity</i>	Win. 2017
<i>Instructor, Physics 231: Methods in Theoretical Physics</i>	Fall 2016
Cornell University, Ithaca, NY	
<i>Teaching Assistant, Physics 3318: Analytical Mechanics</i>	Spr. 2013
<i>Teaching Assistant, Physics 3327: Advanced Electricity & Magnetism</i>	Fall 2012

Course Grader, Physics 7651: Quantum Field Theory I	Fall 2011
UC Berkeley , Berkeley, CA	
Course Grader, Physics 137A: Quantum Mechanics	Sum. 2006
Stanford University , Stanford, CA	
Teaching Assistant, Physics 121: Electrodynamics II	Spr. 2006
Course Grader, Physics 161: Cosmology	Spr. 2006
Course Assistant, The Elementary Particles and their Fundamental Interactions	Sum. 2004

RECENT PRESS

“ Ask a Scientist creates connections and understanding ,” Constance Walter, <i>SURF News and Updates</i>	Aug. 2023
“ Can a theory ever die? ,” Sarah Charley, <i>Symmetry Magazine</i>	May 2022
“ Where do theories come from? ,” Sarah Charley, <i>Symmetry Magazine</i>	Apr. 2022
“ Beyond the Standard Model ,” Katrina Miller, <i>Symmetry Magazine</i>	Feb. 2022
“ What is annihilation? ,” Graycen Wheeler, <i>Symmetry Magazine</i>	Feb. 2022
“ Fourteen UC Riverside professors receive NSF CAREER Awards ,” Holly Ober, <i>Inside UCR</i>	Aug 2021
“ Welcome Back ,” Imran Ghori, <i>UCR Magazine</i>	Sum 2021
“ Billionaires in the great beyond ,” <i>UCR News</i>	July 2021
“ A new dimension in the quest for dark matter ,” Iqbal Pittalwala, <i>UCR News</i>	June 2021
“ Hackathons catch on for creativity, education, and networking ,” Toni Feder <i>Physics Today</i> 74, 5, 23 (2021)	May 2021
“ Veteran and CC transfer student heads to UCLA ,” Iqbal Pittalwala, <i>UCR News</i>	May 2021
“ UCR accepted into Inclusion, Diversity and Equity Alliance ,” Davina Garcia, <i>The Highlander</i>	Nov 2020
“ Physics and Astronomy joins elite group promoting equity, diversity, and inclusion ,” Iqbal Pittalwala, <i>UCR News</i>	Oct. 2020
“ Teaching: Assessment in a Continuing Pandemic ,” Beckie Supiano, <i>The Chronicle of Higher Education</i>	Aug 2020
“ Four professors honored with Senate Faculty Awards ,” Imran Ghori, <i>Inside UCR</i>	June 2020
“ What Goes On in a Proton? Quark Math Still Conflicts With Experiments. ,” Charlie Wood, <i>Quanta Magazine</i>	May 2020
“ Get to know 10 early-career theorists ,” Emily Ayshford <i>Symmetry Magazine</i>	Jul. 2019
“ Massless particles can't be stopped ,” Madeleine O'Keefe, <i>Symmetry Magazine</i>	Jul. 2019
“ How Close Are We to Finding Dark Matter? ,” Paige Keipper, <i>Seeker</i>	Mar. 2019
“ Physics undergraduate student headed to CERN ,” Iqbal Pittalwala, <i>UCR News</i>	Dec. 2018

“ <i>Get to know 10 early-career theorists</i> ,” Emily Ayshford, <i>Symmetry</i>	Jul. 2018
“ <i>Particles crossing to our world could open portal to dark-matter realm</i> ,” Daniel Cossins, <i>New Scientist</i>	Nov. 2018
“ <i>Telling the Truth</i> ,” <i>UCR Magazine</i> ,	Nov. 2018
“ <i>Office Hours: This dark matter expert keeps his office vibe lighthearted</i> ,” Omar Shamout, <i>UCR Magazine</i>	Nov. 2018
“ <i>Dark Matter Eludes Particle Physicists</i> ,” <i>NPR Science Friday</i>	Jun. 2018
“ <i>What’s the Universe Made of?</i> ,” <i>Nova Wonders</i> Season 45 Episode 106	May 2018
“ <i>Nova Wonders Asks ‘What’s the Universe Made Of’ Tonight</i> ,” Sarah Lewin, <i>Space.com</i> & Facebook Live interview	May 2018
“ <i>Nova Wonders Episode to Feature UCR Expert on Dark Matter</i> ,” Iqbal Pittalwala, <i>UCR Today</i>	Apr. 2018
“ <i>Physicists Win Two Awards at Science Hack Day</i> ,” Iqbal Pittalwala, <i>UCR Today</i>	Oct. 2017
“ <i>UCR physicists win ‘Best in Show’ and ‘Most Whimsical’ at Science Hack Day</i> ,” Cydney Contreras, <i>The Highlander</i>	Oct. 2017
“ <i>Generation: First</i> ,” <i>UCR Magazine</i> (Vol 13, No. 1)	Win. 2018
“ <i>Protophobic Fifth-Force Interpretation of the Observed Anomaly in ^8Be</i> ,” Altmetric attention score: 977 Top 5% of all research outputs, #5 output from <i>Phys. Rev. Lett.</i> , 99th percentile for outputs of the same age and source.	Aug. 2016
“ <i>Nuclear Puzzle May Be Clue to Fifth Force</i> ,” Sean Nelson, <i>UCR Today</i>	Aug. 2016
“ <i>Meet Philip “Flip” Tañedo</i> ,” Pinoy Scientists	June 2016
“ <i>One Higgs is the loneliest number</i> ,” Katie Elyce Jones, <i>Symmetry Magazine</i>	July 2015
“ <i>Hard times for theorists in a post-Higgs world</i> ,” Andrew Grant, <i>Science News</i>	June 2013

REFEREEING

Referee: <i>Physics Review D</i> (APS)	2014 – 2023
Referee: <i>Journal of High Energy Physics</i>	2020 – 2022
Referee: <i>Physics Letters B</i>	2022
Referee: <i>European Physical Journal C</i> (Springer)	2014, 2017 – 2018
Referee: <i>Physics of the Dark Universe</i> (Elsevier)	2013, 2015, 2023
Referee: <i>Annals of Physics</i> (Elsevier)	2013
Referee: <i>Europhysics Letters</i> (EPS)	2012

UC REVIEWS

Panelist: UC President’s Postdoctoral Fellowship Program Panel review for Mathematics, Engineering, and Physical Sciences	2023, 25
Off-Site Reviewer: UC President’s Postdoctoral Fellowship Program	2018 – 2021
Reviewer: UC HBCU Initiative In-person review committee for University of California	2023

	Reviewer: UCR Research & Economic Development NSF CAREER internal review Internal feedback for UCR NSF CAREER proposals.	Jun. 2022
GRANT AND TENURE REVIEWS	Panelist, Reviewer: DOE HEP Intensity Frontier University Comparative Review Reviewer: NSERC Discovery Grant Review for the Natural Sciences and Engineering Research Council of Canada.	2021, 2024 2022, 2024
	Reviewer: DOE HEP Theory Comparative Review Mail-in reviewer for the Department of Energy's Office of Science	2023
	Reviewer: Loyola University Chicago Invited confidential evaluation for a promotion to tenure	2023
	Reviewer: Whitman College Faculty Personnel Committee Invited confidential reviewer for a promotion to full professor	2023
PHYSICS SERVICE	Member: High Energy Physics Advisory Panel (HEPAP) Advises the Federal Government on the national program in high energy physics. The panel is chartered by the Department of Energy and reports directly to the Associate Director, Office of High Energy Physics, Office of Science (DOE), and the Assistant Director, Mathematical & Physical Sciences Directorate (NSF)	2023 – 2026
	Chair: APS DPF JJ and Noriko Sakurai Dissertation Committee Vice-chair (2023-24) and then chair (2024-25) of selection committee for the American Physical Society's Division of Particles and Fields	2023 – 2025
	Liaison: Theory/Cosmic Frontier, Snowmass 2022 Community Planning Exercise Decadal study to identify a scientific vision for the future of particle of physics in the US.	2020 – 2022
	Organizing Committee: AIP Communicating Science Workshop	2018
UC SERVICE	Panelist: UC President's Postdoctoral Fellowship Program "Tips for tenure in math, engineering, & physical sciences" for the PFP fall meeting.	2023
UCR SERVICE	Faculty Advisor: Physics Organization for Women and the Underrepresented Faculty advisor for a student group supporting women and underrepresented people in physics.	2023 –
OTHER SERVICE	Finalist Interviewer: Paul & Daisy Soros Fellowships for New Americans Physics Reader: Paul & Daisy Soros Fellowships for New Americans	2020, 2022 2018 – 2021
UNIVERSITY COMMITTEES	Advisory Committee: Advancement and Retention in Academe (ARA) Oversight for a UCR CNAS faculty mentoring program supported by a UCOP Advancing Faculty Diversity grant. PI: Patty Springer, director: Linda Walling	2023 –
	Steering Committee: Science and Technology Studies Initiative One of five steering committee members to develop an interdisciplinary STS program at UCR.	2023 –
	Member: Diversity, Equity & Inclusion Climate Council Committee led by Vice Chancellor and Chief Diversity Officer for campus equity.	2023 –
	Equity Advisor: UCR College of Natural and Agricultural Sciences Works with the Office of DEI and the dean to provide leadership in matters of equity, diversity, and inclusion for faculty recruitment and retention and graduate success.	2022 –
	Committee: UCEAP 3-year Summer Physics Program Review (UC Systemwide)	2019 – 2020
DEPARTMENT COMMITTEES	Fall Chair: Colloquium Committee Coordinating and organizing fall quarter colloquia.	2023 – 2024

Faculty Editor: Hiring Committee: Advancing Faculty Diversity Stepped down due to COVID-19 family care commitments	2019 – 2020
Chair: Website & Display Overhaul and maintenance of department's digital identity.	2017 –
Outreach	2017 – 2018
Graduate Diversity Charged with assessing graduate inclusiveness and making recommendations.	2017 – 2021
Department Climate Committee Charged with assessing graduate inclusiveness and making recommendations. Previously: organized department town halls.	2021 –
Co-Chair: Summer Academy for HS teachers NSF-sponsored program for local physics high school teachers. Designed and initiated equity in physics book club component in 2019.	2016 – 2018
Faculty Hiring Committee: Early Universe Theoretical astrophysics assistant professor search.	2016 – 2017
Chair: Open House Committee Led department participation in Riverside Long Night of Arts & Innovation 2017 and 2019; led department open house in conjunction with UCR Discovery Day in 2016 and 2017.	2016 – 2022
UCR-CC Bridge	2016 – 2017
Ernest Ma Retirement Symposium	2016 – 2017
Undergraduate Advising Included advising 9 undergraduate physics majors during this time.	2016 – 2018

DEPARTMENT SERVICE

Workshop: Writing Graduate School and Fellowship Applications Workshop for undergraduate students applying for graduate school and fellowships.	Oct 2024
<i>“The Secret Syllabus: Unwritten Rules for a Physics Major,”</i>	
Invited talk, Physics 39 (1st year physics majors), UCR	Oct. 2023
Invited talk, Physics 39 (1st year physics majors), UCR	Nov. 2022
Invited talk, Physics 39 (1st year physics majors), UCR	Nov. 2020
Organizer: UCR Physics & Astronomy Spring Town Hall Part of a series of town halls to improve the graduate student climate. This was moved to the auspices of the climate committee.	May 2022
Organizer: UCR Physics & Astronomy Spring Town Hall Part of a series of town halls to improve the graduate student climate. This was moved to the auspices of the climate committee.	May 2022
Organizer: Department Un-Town Hall Small, informal meetings between department stakeholders to improve the department climate.	2022
Speaker: Diversity, Equity, and Inclusion (UCR Physics 401 Professional Development)	Feb 2022
Organizer: APS IDEA: UCR Team Lead American Physical Society network of teams to improve institutional equity.	2020 – 2022
Co-Founder, Advisor: Physics Organization for Women and the Underrepresented Inclusive women-in-physics group to support under-represented students.	2018 –
Social Media: Particle Theory Twitter (@UCR_HEP_Theory)	2017 – 2022
Organizer: UCR Physics Hack Day Weekend student hackathon for digital science projects.	2016 – 2017

UNIVERSITY ACTIVITIES

Workshop: UCR Science to Policy Certificate Program, communicating your science Workshop on effective science communication as an early-career academic	2025, 2022
Faculty Research Video: UCR University Relations Video Department Invited to present a one-minute summary of my research, shared on UCR Communications platforms.	Aug. 2023
Panel: CNAS welcome for new undergraduate students Presentation and panel discussion for incoming first year and transfer students.	Sep. 2022

	Panelist: CNAS/Black Study Initiative Roundtable Invited panel to discuss potential interactions and synergies between faculty, students, and staff in CNAS and the Black Study Initiative at UCR	May 2022
	Moderator: After Grad: Academic Pathways Week Keynote Panel, UCR GradSuccess Invited moderator for a panel of first-generation UC faculty.	Nov. 2021
	Speaker: UCR Science to Policy Certificate Program Led discussion and workshop on social media as an academic.	2018
	Mentor: UCR Graduate Student Mentoring Program Organized by UCR Graduate Division; mentees listed below.	2017 – 2018
	Discussion leader: UCR GradEdge/JumpStart summer journal club Led a journal club to prepare incoming under-represented minority grad students.	2017
	Media contact: UCR Experts on Demand	2016
NON-UNIVERSITY SERVICE	Moderator: Victoria Reyes in conversation with Flip Tanedo Moderator for a discussion with Prof. Victoria Reyes on her new book, <i>Academic Outsider</i> . Hosted by Cellar Door Bookstore.	Oct. 2022
MENTORING	APS National Mentoring Community: Matthew Lugatiman APS National Mentoring Community: Peter Carney Cientifico Latino Mentoring Program: George Iskander APS National Mentoring Community: Sergio Garcia UCR Graduate Student Mentoring Program Pak Kau Lim, Yawei Qin, Joshua Raimist, Lexi Costantino, Nicholas Goman, Yunxiao Li Paul & Daisy Soros Fellowship for New Americans: Laura Cheng	2022 – 2022 – 2019 – 2020 2018 – 2019 2017 – 2018 2017 – 2018
OUTREACH	Ask a Scientist: Sanford Underground Research Facility Public question and answer session at the Sanford Lab Homestake Visitor Center in South Dakota Organizer: Physical & Natural Science Book Club (Cellar Door Bookstore) Monthly discussion of popular science books with members of the community Featured Scientist: STEM Trading Cards , Science Delivered Science Pen-Pal: Letters to a Pre-Scientist program Penl-Pal with a junior high student for one year. Organizer: UCR Summer Physics Teacher Academy, Women In Physics Book Club Adoptee: ΣΠΣ Adopt-A-Physicist program Discussion leader: UC President's Fellowship Spring Retreat; career navigation Science Ambassador: Science Hack Day San Francisco One of six invited scientists sponsored by Gordon & Betty Moore Foundation Blogger, organizer, editor: ParticleBites, a research digest Leads team of students who write about physics for the public. Blogger: US/LHC Blogs / Quantum Diaries, public outreach blog Lecturer: Particle Physics for Undergrads (Cornell undergraduate research program) Lecturer: Stanford ESP SPLASH! and SHEP high school courses on particle physics	2023 2016 – 2020 – 2021 2019 – 2020 2019 – 2020 2018 – 2019 2017 2015 2013 – 2009 – 2012 2012 2005
PUBLIC TALKS & PANELS	"Scotohylogy," <i>Ologies</i> podcast interview by Alie Ward; rebroadcast on RadioLab "Little Particles in a Big Universe," virtual talk at FilSciHub, Philippines (remote) "Year of ParticleBites," virtual talk at Sustainable HEP 2nd ed., CERN (remote) "Explainer Videos: scalable, inclusive, and metacognitive assessment," Contributed talk at the APS April Meeting 2022, New York, NY	Feb. 2023 Oct. 2022 Sep. 2022 Apr. 2022

“Pursuing Diversity and Inclusion as an Early Career Scientist,” Invited talk at the APS April Meeting 2022, New York, NY	Apr. 2022
“ <i>Popular book discussions: a platform for equity in physics</i> ,” doi:10.26081/K6J330 KITP Teachers’ Conference: Neutrinos and New Physics, Santa Barbara, CA	Mar. 2022
“#PhysicsMeetup: <i>Little Particles in a Big Universe</i> ,” (video) ICTP Physics Without Frontiers: Philippines	Jul. 2021
Panel: “ <i>Physics Education</i> ,” UCR Summer Physics Teacher Academy	Jun. 2021
“ <i>Non-Traditional Assessment & Inclusivity in Physics</i> ,” UCR Summer Physics Teacher Academy	Jun. 2021
“ <i>Radio Physics</i> ,” Invited interviewee, KDNK Community Radio, Aspen, CO Panel organizer: “ <i>The Only Woman in the Room</i> book club discussion” UCR Summer Physics Teacher Academy	Jun. 2019 2019, 2022
“ <i>Finding Your Voice: Storytelling Lessons for Scientists</i> ” 2019 AAAS Meeting, invited panel from Science Friday; Washington DC	Feb. 2019
“ <i>Attending to Diversity In the Physics Classroom</i> ” UCR Summer Physics Teacher Academy Conference	Feb. 2019
“ <i>How to Fail as a Physics Major</i> ,” Invited talk, Physics 39 (1st year physics majors), UCR Invited talk, Physics 39 (1st year physics majors), UCR	Nov. 2019 Nov. 2018
“ <i>Impostor Syndrome and Failure in College</i> ” UCR Society of Physics Students	Oct. 2018
“ <i>My New American Story</i> ” Paul & Daisy Soros Fellowships for New Americans Fall Conference Story Slam	Oct. 2018
“ <i>The Physics of Star Wars</i> ” UCR ARTSblock film screening and discussion	May 2018
“ <i>Breaking into Academia</i> ” UCR TRIO Scholars Program	Apr. 2018
“ <i>First-Gen Day Faculty Panel</i> ” UCR First Generation Friday Panel	Jan. 2018
“ <i>Surviving the Academic Job Market</i> ” UCR Grad Success Panel	Nov. 2017
“ <i>First Generation Academics</i> ” UCR Grad Success Panel	Nov. 2017
“ <i>I have no idea: A guide to finding your way in science</i> ” Invited talk, Physics 39 (1st year physics majors), UCR Middle School Symposium talk, Riverside STEM Academy High School Symposium talk, Riverside STEM Academy	Nov. 2020 Sep. 2017 Sep. 2017
“ <i>Quantum Electrodynamics with Feynman Diagrams</i> ” UCR Summer Physics Teacher Academy	Jun. 2017
“ <i>Applying to Graduate School</i> ” UCR Society of Physics Students	May 2017
“ <i>Communicating through Social Media</i> ” Communicating Science: San Diego, Invited Panelist	Sep. 2016
“ <i>Data vs. Dark Matter: Opportunities for Open Science in Particle Physics</i> ” Science Hack Day, San Francisco (Gordon & Betty Moore Science Ambassador)	Oct. 2015
“ <i>Balancing writing your dissertation and finding a postdoc position</i> ” Panelist, UCI Graduate Division & ADVANCE Program	Nov. 2014
“ <i>Here be dragons: [fairy] tales from beyond the Standard Model</i> ” Cornell Society of Physics Students	Apr. 2013
“ <i>The Physics of Angels & Demons</i> ” Cornell Institute for Physics Teachers	Jul. 2009
“ <i>Applying for Graduate Fellowships Information Session</i> ”	

	Panelist, Cornell Physics Department	Oct. 2008
	<i>“Quantum penguins hint at the nature of matter and antimatter”</i>	
	Marshall Aid Commemoration Commission Colloquium, London, UK	May 2008
PHD STUDENTS	Robert Clemenson , Visiting Ph.D student from Sussex (Advisor: Stephan Huber)	2022
	Yash Aggarwal , Ph.D student, UC Riverside (co-advisor: Anson D’Aloisio)	2019 –
	Alexandria Costantino , Ph.D student, UC Riverside	2018 – 2022
	Adam Green , Ph.D student, UC Riverside	2018 –
	Kuntal Pal , Ph.D student, UC Riverside (co-advisor: Jose Wudka)	2018 – 2023
	Ian Chaffey , Ph.D student, UC Riverside	2017 – 2022
POSTDOCTORAL SCHOLARS	Daneng Yang , Postdoctoral Scholar, UC Riverside (primary mentor: Hai-Bo Yu)	2021 –
	Aniket Joglekar , Postdoctoral Scholar, UC Riverside (primary mentor: Yanou Cui)	2017 – 2020
	Ran Huo , Postdoctoral Scholar, UC Riverside	2016 – 2018
UNDERGRADUATE STUDENTS	Matthew Lugatiman , undergraduate researcher, UC Riverside	2022 –
	Cecelia Ngo , undergraduate researcher, UC Riverside	2021 – 2022
	Sagada Penano (Stanford University), Summer Research Student	2020 – 2021
	Sergio Garcia , APS National Mentoring Community, UC Riverside	2018 – 2019
	Syris Norelli , undergraduate researcher, UC Riverside	2017 – 2018
	Adam Green , undergraduate researcher/honors advisee, UC Riverside	2016 – 2017
HS STUDENTS	Nathan Kang (North Hollywood High School), Research Student	2022 –
	Anagha Satish (Redlands High School), Summer Research Student	2020 – 2021
QUALIFYING EXAM	Non-Chair member of the UCR qualifying exam committee for the following students	
	Sally (Fengyi) Li (Physics)	May 2024
	Rahul Rajkumar (Mathematics)	Jan. 2024
	Jialin Wang (Mathematics)	May 2023
	Elliott Vest (Mathematics)	Nov. 2022
	Erik Lloyd	Oct. 2021
	Gerardo Alvarez	Nov. 2019
	Andrew Birkey (Earth Sciences)	Oct. 2019
	Michael Shamma	Nov. 2019
	Amartya Mitra	Sep. 2017
	Lydia Elias	Sep. 2017
	Tao Ren	Sep. 2016
PHD DEFENSE	Non-Chair member of the UCR dissertation committee for the following students	
	Gerardo Alvarez	Sep. 2021
	Michael Shamma	Oct. 2019
	Tao Ren	Sep. 2019
	Mohammareza Zakeri	Jun. 2017